Lesson 3.1 - Statistics Formulas and Equations

> **Video:** Statistics - Formulas and Equations by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

A reference sheet of the basic formulas used at the start of a statistics course. Later lessons in this chapter work through each one in detail.

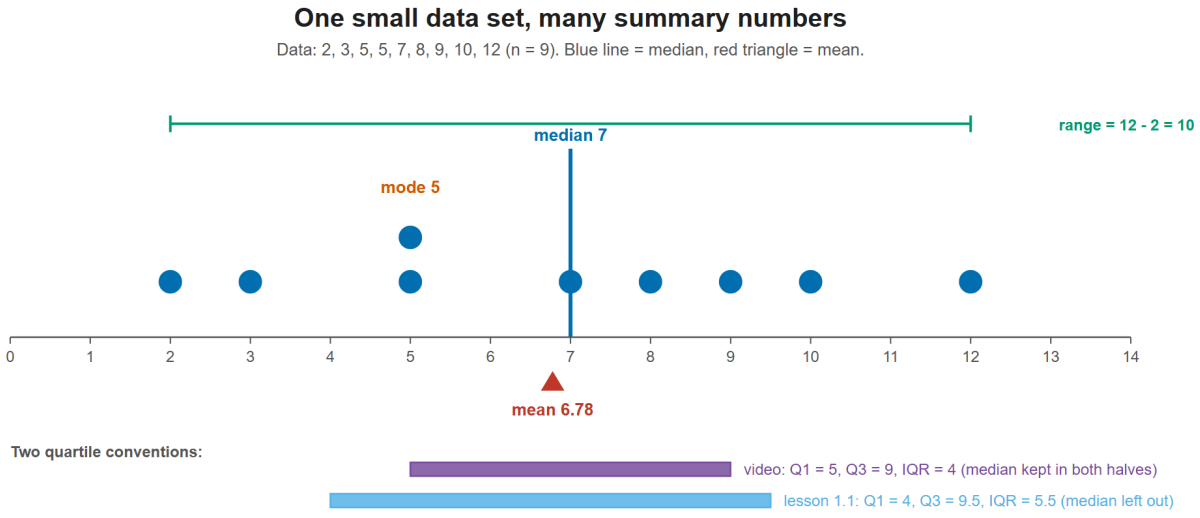
---

1. Median, mode, range, quartiles

Example data ($n = 9$, already sorted): 2, 3, 5, 5, 7, 8, 9, 10, 12

Quantity	Definition	Value
Median (Q_2, 50th percentile)	Middle value	7
Mode	Most frequent value	5 (appears twice)
Range	max − min	$12 - 2 = 10$
Q_1 (25th percentile)	Median of the lower half	5 (video)
Q_3 (75th percentile)	Median of the upper half	9 (video)
IQR	$Q_3 - Q_1$	$9 - 5 = 4$

> **Quartile conventions differ.** The video's $Q_1 = 5$ and $Q_3 = 9$ come from **including** the median 7 in both halves (2 3 5 **5** 7 and 7 8 **9** 10 12). The Lesson 1.1 method **excludes** the median, giving halves 2 3 5 5 and 8 9 10 12, so $Q_1 = 4$ and $Q_3 = 9.5$. Use whichever method your course uses, and use it consistently.



The example data as a dot plot with the median (7), mode (5), mean (6.78) and range (10) marked. The two bars at the bottom show why quartiles can differ between textbooks: keeping the median in both halves gives Q1 = 5 and Q3 = 9, while leaving it out gives Q1 = 4 and Q3 = 9.5.

---

2. Arithmetic (sample) mean

$$\bar{x} = \frac{\sum x}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

Example: $\bar{x} = \dfrac{2+3+5+5+7+8+9+10+12}{9} = \dfrac{61}{9} \approx 6.78$

---

3. Four kinds of mean

For two numbers a and b

Mean	Formula
Arithmetic (AM)	$\dfrac{a+b}{2}$
Geometric (GM)	$\sqrt{ab}$
Harmonic (HM)	$\dfrac{2}{\frac{1}{a}+\frac{1}{b}} = \dfrac{2ab}{a+b}$
Root mean square (RMS)	$\sqrt{\dfrac{a^2+b^2}{2}}$

For any two positive numbers:

$$\mathrm{HM} \le \mathrm{GM} \le \mathrm{AM} \le \mathrm{RMS}, \qquad \mathrm{GM} = \sqrt{\mathrm{AM} \cdot \mathrm{HM}}$$

Quick check with $a = 4$, $b = 9$: AM = 6.5, GM = 6, HM = $72/13 \approx 5.54$, RMS = $\sqrt{48.5} \approx 6.96$. The order is 5.54 ≤ 6 ≤ 6.5 ≤ 6.96 ✓, and $\sqrt{6.5 \times 5.54} = 6$ ✓.

For n numbers

Mean	Formula
Arithmetic	$\bar{x} = \dfrac{x_1 + x_2 + \cdots + x_n}{n}$
Geometric	$(x_1 \cdot x_2 \cdots x_n)^{1/n}$
Harmonic	$\dfrac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}}$
Root mean square	$\sqrt{\dfrac{x_1^2 + x_2^2 + \cdots + x_n^2}{n}}$
Weighted	$\bar{x}_w = \dfrac{\sum wx}{\sum w} = \dfrac{w_1x_1 + w_2x_2 + \cdots + w_nx_n}{w_1 + w_2 + \cdots + w_n}$

With $n = 2$ the geometric-mean formula becomes $(ab)^{1/2} = \sqrt{ab}$.

---

4. Sample vs population symbols

	Sample	Population
Mean	$\bar{x}$	μ
Size	n	N
Standard deviation	s	σ
Variance	s^2	σ^2

Example: a city has $N = 10{,}000$ residents, but a study surveys $n = 300$ of them. n and N are different numbers.

---

5. Standard deviation and variance

	Sample	Population
Standard deviation	$s = \sqrt{\frac{\sum(x-\bar{x})^2}{n-1}}$	$\sigma = \sqrt{\frac{\sum(x-\mu)^2}{N}}$
Variance	$s^2 = \frac{\sum(x-\bar{x})^2}{n-1}$	$\sigma^2 = \frac{\sum(x-\mu)^2}{N}$

- The **sample** formulas divide by $n-1$. The **population** formulas divide by N.
- Variance = (standard deviation)^2, the same formula without the square root.

---

6. Coefficient of variation (CV)

$$\mathrm{CV}_{\text{sample}} = \frac{s}{\bar{x}} \times 100\%, \qquad \mathrm{CV}_{\text{population}} = \frac{\sigma}{\mu} \times 100\%$$

The CV expresses the standard deviation as a **percentage of the mean**. A CV of 0.25 (25%) means the standard deviation is 25% of the mean.

---

7. Mean absolute deviation (MAD)

$$\mathrm{MAD} = \frac{\sum |x - \bar{x}|}{n}$$

It looks like the standard deviation, but it uses **absolute values instead of squares** and has **no square root**.

Key takeaways

- Know the four means (AM, GM, HM, RMS), their order HM ≤ GM ≤ AM ≤ RMS, and the weighted mean.
- Sample formulas use $\bar{x}$, n, s and divide variance by $n-1$. Population formulas use μ, N, σ and divide by N.
- CV = SD ÷ mean × 100%. MAD = average absolute distance from the mean.